Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/14/2021
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Assessment of Nutritional Value and Growth Parameters of Space-Grown Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Space Biology Unsolicited
Start Date:	08/01/2017	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC17K0344		
Performance Goal No.:			
Performance Goal Text:			
	The goal of the proposed work is to determine the effects of space flight on growth, performance (productivity and time to maturation), metabolism, and genetic changes in radish plants (Raphanus sativus). Plants will be grown in the ADVANCED PLANT HABITAT (APH) hardware on the International Space Station (ISS) and in comparable experiments in ground controls. The research conforms to NASA's top future goal of long-term space exploration and terraforming other planets and satellites. These goals depend on nutritionally valuable food crops that can be reliably grown in an environment that is not only defined by altered gravity or weightless environment but also different atmospheric conditions. In contrast to studying direct effects of gravity on plants, our approach focuses on secondary, derived conditions such as altered convection, transpiration, watering conditions, CO2, other gaseous components, and LED illumination, which creates a characteristic energy (heat and light) gradient within the growth volume.		

Task Description:	Radishes are good candidate plants to grow on the ISS because they mature quickly (harvest after = 28 days after planting), have been grown successfully on the ISS (Massa et al., 2013); genetic and metabolic information is readily available (Shen et al., 2013); and they are nutritionally desirable and provide a wealth of information that can thoroughly be analyzed by biochemical, physiological, and genetic approaches. Radishes are good sources of minerals (especially K, Ca, Fe, and Zn), complex carbohydrates, vitamin C, dietary fibers, folic acid (vitamin B9), anthocyanins, and glucosinolates, a class of secondary metabolites characteristic for Brassicaceae. In addition to their roots, certain varieties of radish are used for their leaves, siliques, and oil. The nutritional value and versatility of these vegetables makes them good candidates as a crop plant for astronauts. The small seed size (<4 mm), longevity of seeds (up to five years), small plant size, and the ability to consume the entire plant makes radishes ideally suited for space biology. Previous studies have focused on the effect of gravity and shown that space conditions allow successful cultivation of Brassica species (Musgrave et al., 2000). However, the quality, biomass, sensitivity of growth to day length, and (constant) temperature are important parameters for this species that have yet to be studied, especially in light of results that seed development in space cultures was compromised (Kuang et al., 2005), especially when oxygen was limited (Kuang et al., 1998). Equally unknown is the effect of environmental conditions in spacecrafts on the performance of plants. Understandably, the emphasis on space biology focuses on the effect of weightlessness and gravity on plant orientation and mechanical properties (Hoson 1999). However, conditions that are the consequence of weightlessness, e.g., lack of convection or altered atmosphere, are equally significant but unknown. Our proposal intends to measure the metabolic, physiological, and genetic responses of radishes grown on the ISS and identify the effect of the space environment on metabolite accumulation, flavor, enzyme activity, mineral uptake, and generation time. This knowledge will be essential for the transition from Earth-bound cultivation of plants to growth conditions in space; the research is important for the establishment of human colonies on the Moon, Mars, and farther outposts. References Hoson T (1999) Interaction of gravity with other environmental factors in growth and development: An introduction. In: Duke PJ, Barlow PW, Hoson T, Mogami Y (eds) Life Sciences: Microgravity Research I, pp 1971-1974 Kuang A, Crispi M, Musgrave ME (1998) Control of seed development in Arabidopsis thaliana by atmospheric oxygen. Plant Cell and Environment 21: 71-78 Kuang A, Popova A, McClure G, Musgrave ME (2005) Dynamics of storage reserve deposition during Brassica rapa L. pollen and seed development in microgravity. Int J Plant Sciences 166: 85-96 Massa G, Newsham G, Hummerick ME, Caro JL, Stutte GW, Morrow RC, Wheeler RM (2013) Preliminary Species and Media Selection for the VEGGIE Space Hardware. Grav Space Res 1: 95-106 Musgrave M, Kuang A, Xiao Y, Stout S, Bingham G, Briarty L, Levinskikh M, Sychev V, Podolski I (2000) Gravity independence of seed-to-seed cycling in Brassica rapa. Planta 210: 400-406 Shen D, Sun H, Huang M, Zheng Y, Li X, Fei Z (2013) RadishBase: A Database for Genomics and Genetics of Radish. Plant Cell Physiol 54: E3-+
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objective to grow plants under space conditions requires optimization of auxiliary parameters such as light intensity, quality, watering, substrate definition, fertilization that all have immediate implications for more efficient agricultural practices. The data that we will obtain examine much more than simple yield (mass) of the plants but include the mineral content, propensity to flower, flavor components, and water use. All attempts will not only optimize plant culture under elevated CO2, limited water, and controlled fertilizer provision, but also investigate the genetic responses of plants to mitigate space (or agricultural) stress. The overall result of the project will therefore benefit our generic understanding of factors that control plant growth and productivity in space and on Earth.
Task Progress:	Based on the prior activity (previously submitted as Second year report), activities progressed on two fronts: At the Kennedy Space Center (KSC), an initial cultivation period was followed by the science verification test (SVT1, 5/16 – 6/13/19) and a subsequent Δ -SVT (9/30 – 10/30/19). The current report includes the experiment verification test (EVT), completed 12/16/19, an additional test in the Engineering Demonstration Unit (EDU) unit (4/7 – 5/4/20), and the Δ EVT (5/26 -6/22/20). At the home institution, radish cultivation methods and light management were studied to supplement KSC activities. Experiment Verification Test (EVT): The objective of the EVT was to examine the growth performance of radish seedlings in the Advanced Plant Habitat (APH) in preparation for a planned space experiment under conditions similar to those on the International Space Station by controlling the experiment mostly by remote (indirect) control via the APH software. The test further focused on the most sensitive issues namely water, temperature, and light control of the system. Samples were harvested at KSC and transported in insulating boxes containing dry ice directly to UL Lafayette. APH performance and Plant Development: A substantial improvement to prior science verification tests was a consistent water delivery system. Test in the Engineering Demonstration Unit: The purpose of the EDU test was to compare different fertilizer application (Nutricote vs. ½ strength MS medium) and automated water delivery vs. constant water supply through a ‘stand-pipe’ controlled system. The EDU chamber does not provide the same fidelity as the actual, sealed APH system and therefore the CO2 concentration could not be maintained at the desired 3500 ppm level but at 2000 ppm. Δ -EVT (2. Experimental Verification Test): The objective of the second Experiment Verification Test was to assess the outcome of several adjustments made based on the results of the first EVT (11/20, through 12/16, 2019) and an additional grow-out in the EDU (4/7 – 5/4/20). Primary objectives included improved radish-bulb formation, assessment of the growth patterns, calculation of a modified Normalized Difference Vegetation Index (NDVI), and an examination

	of the harvested tissue in preparation for a planned space experiment. Plant development: Plants were thinned to one seedling per growth position on Day 8 (6/3/20). Harvesting of tissue was performed on June 22, 2020. Despite the reduced number of plants, the overlapping leaves likely reduce light availability and thus photosynthetic performance. The biomass at harvest is likely reduced as a consequence of light competition. The curly appearance of the leaves indicates a stress response that was not observed during EVT and EDU tests. The most likely factor for this observation is the elevated CO2 concentration. Summary: This report of the last three tests in preparation of the flight experiment (scheduled for 9/29/2020) indicates that the growth conditions and results have been substantially improved. The watering system is functioning and the adjustment in the lighting regime (enhanced blue light) promoted bulb formation. The only parameter that cannot be compared with either EVT or EDU data is the elevated CO2 concentration during Δ -EVT. Morphological changes (curling leaf edges and necrosis) indicate high CO2 stress levels. The mineral and nitrogen analyses suggest that enhanced P/K/N provision such as full-strength MS medium as opposed to the half-strength medium used in the Δ -EVT may mitigate these phenomena. The biological material is sufficient to assess the metabolism, morphology, and transcriptional activity after cultivation in the Advanced Plant Habitat. Lastly, Solid Phase Gene Extraction (SPGE) sampling is a promising novel technology for space applications and allows for non-destructive transcriptional assessments.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Articles in Other Journals or Periodicals	Pesacreta TC, Acharya A, Hasenstein KH. "Distribution and concentration of endogenously derived elements in Zea mays (corn) seeds and seedling root tissues." Protoplasma.in press as of January 2021. , Jan-2021
Articles in Peer-reviewed Journals	Kiss JZ, Wolverton C, Wyatt SE, Hasenstein KH, van Loon JJWA. "Comparison of microgravity analogs to spaceflight in studies of plant growth and development. " Front Plant Sci. 2019 Dec 6;10:1577. Review. https://doi.org/10.3389/fpls.2019.01577 ; PubMed PMID: 31867033; PubMed Central PMCID: PMC6908503 , Dec-2019
Articles in Peer-reviewed Journals	John SP, Hasenstein KH. "Desiccation mitigates heat stress in the resurrection fern, Pleopeltis polypodioides." Front Plant Sci. 2020 Nov 30;11:597731. https://doi.org/10.3389/fpls.2020.597731 ; PMID: 33329661; PMCID: PMC7733933 , Nov-2020